Birla Institute of Technology & Science, Pilani

Online B.Sc. Computer Science Programme
Second Semester 2022-2023
Comprehensive Exam
(Slot-2)

Course No.	: BCS ZC233
Course Title	: Probability & Statistics
Nature of Exam	: Open Book
Weightage	: 50%
Duration	: 2 Hours 30 Minutes
Date of Exam	: 13th October 2023

No. of Pages = 2
No. of Questions = 5

	Question	Marks
Q.1.a	i).If A and B are mutually exclusive and P (A) = 0.29, P (B) = 0.43. Find 1). $P(A \cap \bar{B})$ $P(A \cap \bar{B})$ 2) P $(\bar{A} \cap \bar{B})$ $(\bar{A} \cap \bar{B})$ 3) P $(\bar{A} \cap B)$ $(\bar{A} \cap B)$ ii) If P(A)=1/3 , P(B)=1/2 , P(A/B) = 1/6 find P(B/A)	**5 M**
Q.1.b	Consider the following data. A: the event that a patient selected is with high B.P B: the event that a patient selected is diabetic C: the event that a patient selected is with cancer	**5 M**

Events	A	B	C	A and B	B and C	A and C	(A and B and C)
Probability	0.30	0.30	0.28	0.15	0.10	0.20	0.12

(i). If a patient is selected at random, what is the probability that he suffering none of these diseases?

(ii). What is the probability that the patient selected is with B.P given that he is with either of the remaining two diseases?

(iii). What is the probability that the patient is diabetic given that he is suffering from at least one of the diseases mentioned?

	Question	Marks
Q.2.	A random variable X X has the following probability function.	**10 M**

X	0	1	2	3	4	5	6	7
$P(X)$	0	$2k$	k	k	$2k$	k^2+k	$2k^2$	k^2

(i) Find k k

(ii) Evaluate $P(X<6)$ $P(X<6)$, $P(X \geq 6)$ $P(X \geq 6)$, $P(0<X<5)$ $P(0<X<5)$, $P(5<X \leq 7)$ $P(5<X \leq 7)$

(iii) Mean

(iv) Variance

	Question	Marks
Q.3.a	A random variable follows binomial distribution with n = 400 and p = 0.01, then find	**5 M**

<table>
<tr><td></td><td>$P(X > 175)$</td><td></td></tr>
<tr><td>Q.3.b</td><td>The following are the weights in decagrams of 10 packages of grass seed distributed by a company 46.4, 46.1, 45.8, 47, 46.1, 45.9, 45.8, 46.9, 45.2 and 46. Find 99% confidence interval for the variance of all such grass seed distributed by the company assuming normal distribution.</td><td>5 M</td></tr>
<tr><td>Q.4.a</td><td>A sample of 100 electric bulbs produced by company A has mean lifetime of 1190 hours and a standard deviation of 90 hours. A sample of 75 bulbs produced by company B has mean lifetime of 1230 hours with a standard deviation of 120 hours. Is there a difference between the mean lifetime of two brands at a significant level of 0.05?</td><td>5 M</td></tr>
<tr><td>Q.4.b</td><td>A Random sample of 100 college students weighed on average 63 kg with a standard deviation of 10 kg. Test the hypothesis that $\mu=60$ kg against the alternative $\mu < 60$ at 0.05 level of significance.</td><td>5 M</td></tr>
<tr><td>Q.5.a</td><td>Check whether the following variables are having linear correlation or not. Comment on the strength of the relation if exists.
<table>
<tr><td>X</td><td>25</td><td>54</td><td>35</td><td>20</td><td>18</td></tr>
<tr><td>Y</td><td>18</td><td>45</td><td>30</td><td>18</td><td>12</td></tr>
</table></td><td>5 M</td></tr>
<tr><td>Q.5.b</td><td>Fit simple linear regression $y = a + bx$ on the following data.
<table>
<tr><td>X</td><td>10</td><td>15</td><td>18</td><td>16</td><td>14</td></tr>
<tr><td>Y</td><td>12</td><td>14</td><td>10</td><td>12</td><td>15</td></tr>
</table></td><td>5 M</td></tr>
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